

Lesson 10

Distribution of Primes & their properties

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Objectives

At the end of this session you should be able to

- Define the prime numbers and composite numbers
- Prove, there are infinitely many primes
- Prove the Fundamental Theorem of Arithmetic
- Use the Sieve of Eratosthenes to check whether a given integer is prime
- Use the Sieve of Eratosthenes to represent a given integer as a product of prime factors.

10.1 Introduction

Among all the positive integers 1, 2, 3, 4, ... we can find that some integers have only two distinct positive divisors and the others have more than two positive divisors, except the integer 1, which has just one positive divisor, namely itself.

First ten numbers which have only two distinct positive divisors are 2, 3, 5, 7, 11, 13, 17, 19, 23, and 29. Observe that only divisors of each of these numbers are 1 and itself. These numbers are called prime numbers. The interesting thing about prime numbers is we cannot see any pattern on their distribution. However there are infinitely many prime numbers. The proof goes back to the ancient Greek period. Also any positive integer greater than 1, which is not prime, can be written

as product of primes (allowing the repetition) in a unique way. This theorem is very important in number theory, called the Fundamental Theorem of Arithmetic.

There are many unsolved problems related to the prime numbers. Gold Bach conjecture is a prominent one. It says that every even number greater than two can be written as sum of two primes.

10.2 Properties of primes

10.2.1 Definition

An integer $p \geq 2$ is prime if it has no positive divisors other than 1 and itself. An integer greater than or equal to 2, which is not prime, is composite.

Note that 1 is neither prime nor composite.

For example 2, 3, 5, 7, 11,are prime, but 4, 6, 8, 9, 10are composite. Among all the even integers, only 2 is prime, and all others are composite.

10.2.2 Theorem

Every integer greater than 1 is prime or can be expressed as a product of primes.

Proof:

Let n be any integer greater than 1.

If n is prime, nothing to prove. Suppose n is not prime.

Then $n = n_1 n_2$ for some $1 < n_1 < n$ and $1 < n_2 < n$ $\{n_1, n_2$ need not necessarily be distinct $\}$.

If n_1 is prime let it stand, otherwise $n_1 = n_3 n_4$ for some $1 < n_3 < n_1$ and $1 < n_4 < n_1$.

Similar argument hold for n_2 .

If proceed in this manner, product of factors must be terminate because the factors are smaller than n and each factor is greater than 1.

Hence we can write n as a product of primes.

Since the prime factors are not necessarily distinct, the result can be written in the form

$n = p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}$. Where $p_1, p_2, \dots, p_r$ are distinct primes and $a_1, a_2, \dots, a_r$ are positive integers. ■

10.2.3 Theorem

If p is a prime and $p|ab$, then $p|a$ or $p|b$.

Proof:

If $p|a$ nothing to prove. Suppose $p \nmid a$. Since only positive divisors of p are 1 and p itself, $\gcd(a, b) = 1$. Hence by Euclid's lemma $p|b$. ■

The following corollary generalizes the above theorem for any finite product of integers.

10.2.4 Corollary

If p is a prime and $p|a_1 a_2 \dots a_k$, where k is a positive integer, then $p|a_1$ or $p|a_2$ or ... or $p|a_k$.

Proof : Use the induction and above theorem. We left the proof for the reader.

10.2.5 Theorem- Euclid's theorem

There are infinitely many primes.

Proof:

Suppose the number of primes is finite. Hence the last prime exists. Let it be p_k .

Let $p_1 = 2, p_2 = 3, \dots$ and so on.

Let $N = p_1 p_2 \dots p_k + 1$. Note that N is not divisible by p_1 or p_2 or ... or p_k .

Hence any prime divisor p of N is different from $p_1, p_2, \dots, p_k$.

Since N is either prime or has a prime factor, this implies that there is a prime distinct from $p_1, p_2, \dots, p_k$. This contradicts with list of all primes are $p_1, p_2, \dots, p_k$.

Hence the number of primes is infinite. ■

10.2.6 Theorem -The Fundamental Theorem of Arithmetic

The factoring of any integer $n > 1$ in to primes is unique apart from the order of the prime factor.

Proof:

Suppose that there is an integer N which has two different prime factoring.

Now by dividing out any primes common to the two representations, we ended up with

$p_1 p_2 \dots p_r = q_1 q_2 \dots q_s$ where p_i s and q_i s are primes, not necessarily all distinct.

But no primes in the left side occur on the right side.

Now $p_1 p_2 \dots p_r = q_1 q_2 \dots q_s$ implies $p_1 | q_1 q_2 \dots q_s$.

By Corollary 10.2.4, p_1 is a divisor of at least one of the q_i .

Hence $p_1 = 1$ or $p_1 = q_i$ for some $i = 1, 2, \dots s$.

This is a contradiction. Hence the result. ■

10.2.7 Definition

Prime numbers p and q such that $p < q$ is said to be twin primes if $q = p + 2$.

Activity 01

(i) Show that a number is a square if and only if all the exponents in its prime factorization are even.

(ii) If p is a prime number, determine all the divisors of p^n .

(iii) Show that any positive integer n can be written uniquely as $n = 2^r s$, where s is odd and r is a non-negative integer. Express the integers from 5 to 100 in this form.

10.3 Sieve of Eratosthenes

Given a positive integer, how we determine whether it is prime or composite? In the later case how we find all the divisors of it?

The most trivial approach consists of successively dividing that integer from each of the positive integers preceding it. But do we need to divide from all the preceding positive integers? Closer look suggest that dividing by all the preceding primes is sufficient.

Can we reduce this process furthermore? Let that positive integer be M . If it is divisible by a prime, greater than $\sqrt{M}$, then all other factors must be less than $\sqrt{M}$. This suggest that, to find all the prime divisors it is sufficient to divide M by primes not exceeding $\sqrt{M}$. This procedure is called the Sieve of Eratosthenes.

10.3.1 Example

Determine whether 509 is prime or composite.

Answer:

Observe that $22 < \sqrt{509} < 23$. Hence we need only try out the primes that are not larger than 22, namely the primes 2, 3, 5, 7, 11, 13, 17, 19. Each of these primes are not a divisor of 509. Hence 509 is prime.

10.3.2 Example

Determine whether 2093 is prime or composite. If it is composite write down it as a product of primes.

Answer:

Observe that $45 < \sqrt{2093} < 46$. Hence we need only try out the primes that are not larger than 45, namely the primes 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43. By trial the first prime divide 2093 is 7. Also $2093 = 7 \cdot 299$. Now Observe that $17 < \sqrt{299} < 18$, hence find the factors of 299 try out the primes 2, 3, 5, 7, 11, 13, 17. The first prime divide 299 is 13, carrying out the required division we obtain $299 = 13 \cdot 23$. But 23 is itself a prime. Hence $2093 = 7 \cdot 13 \cdot 23$.

Activity 02

- (i) Determine whether the integer 701 is prime, by using the Sieve of Eratosthenes.
 - (ii) Prove that any composite three digit number must have a prime factor less than or equal to 31 .
 - (iii) Prove that any prime number greater than 3, is of the form $6k - 1$ or $6k + 1$ for some positive integer k . Deduce that if p is a prime and $p > 3$ then 24 is a factor of $p^2 - 1$.
 - (iv) Prove that, if p_n is the n^{th} prime number, then $p_n \leq 2^{2^{n-1}}$
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